# HARSHA LOLEWAR

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## PROFILE

BSc Computer Science student specializing in Cognitive Systems with an 8.1 CGPA, seeking internship opportunities in AI/ML, software development, and related computing fields. Has shipped live web apps, including a real-time multiplayer game, an AI debugging assistant, a 3D Solar System explorer and a Raft fault simulator, alongside a foundation in Python, Java, C, SQL, prompt engineering, and Git/GitHub.

## EDUCATION

**Loyola Academy**, Hyderabad — **2025–2028**
BSc Computer Science — Specialization: Cognitive Systems — **CGPA: 8.1/10**

## TECHNICAL SKILLS

| | |
|---|---|
| **Programming** | Python, Java, C, SQL, JavaScript |
| **Web** | React, WebSockets, Three.js (React Three Fiber), GLSL shaders, Cloudflare Workers & Pages |
| **AI / ML** | Prompt Engineering, Google AI Studio (Gemini), AI-assisted development |
| **Systems** | Raft consensus, discrete-event simulation, Pyodide (WebAssembly) |
| **Databases** | MySQL, IndexedDB |
| **Tools** | Git, GitHub, GitHub Actions, VS Code |

## PROJECTS

**Sync or Solo — Real-time Multiplayer Party Game** (Live · Code)

- Browser party game for 2–8 players on their phones: everyone answers the same prompt, then bets on matching the crowd (Sync) or standing alone (Solo).
- Built with React and WebSockets for live room-based play with 4-letter room codes; deployed on Cloudflare Workers. No accounts or installs; idle rooms expire after 30 minutes.

**BugLoop — AI Debugging Knowledge Assistant** (Live)

- AI debugging workbench for student developers, vibe-coded in Google AI Studio for the Handshake AI Skill Studio mission.
- Diagnoses code errors step by step, saves confirmed fixes to a knowledge base stored locally in the browser (IndexedDB), and turns repeated error patterns into reusable debugging rules.

**Solar System — Interactive 3D Solar System Explorer** (Live · Code)

- 3D Solar System with every planet at its real position for today (JPL orbital elements) and Earth turning with the real clock.
- Fly to the Sun, 8 planets and 13 moons, visit surface landmarks, read NASA fact sheets; React, Three.js (R3F), GLSL, Zustand.

**Raft Failure Lab — Distributed-Systems Fault Simulator** (Live · Code)

- Deterministic simulator of a 5-node Raft cluster: inject delays, drops, duplicates, crashes and partitions; runs repeat by seed.
- Invariant checker verifies Raft's safety guarantees after every run, pointing to the exact event of any violation; timeline replay.
- Pure Python in the browser via Pyodide (WebAssembly); test-first unittest suite; GitHub Actions CI deploys to Cloudflare Pages.

## ACHIEVEMENTS & HACKATHONS

- **1st Prize — Brain Rush & Technoverse, TECHXCELERATE, Loyola Academy (2026):** Secured first place in a competitive event recognizing knowledge, problem-solving ability, and competitive spirit.
- **SPEC'S Industry Hack 2026 — National-Level Hackathon:** Participated in a 24-hour national-level hackathon at St. Peter's Engineering College as part of a 2-member team; contributed to frontend development and bug fixing.
- **2-Day State-Level Hackathon Challenge, Loyola Academy (2026):** Participated in the state-level hackathon conducted on 8–9 January 2026.
- **Handshake AI Skill Studio (2026):** Completed missions including vibe-coding BugLoop and designing a Halloween-themed logo (view).

## AI / ML PROGRAM

**Specialized Program in Artificial Intelligence & Machine Learning with Drone Tech — TiHAN, IIT Hyderabad × Masai:** Enrolled in the specialized online program beginning July 2025, focused on AI/ML with Drone Tech.

## RESEARCH

**Co-author — GroundRAG Research Paper:** Contributed to a research paper on GroundRAG. The paper is currently unpublished; publication details are not yet available.

## EXTRACURRICULARS & LANGUAGES

- **Interests:** Guitar, fitness / gym
- **Campus involvement:** Member of the Music Club during first semester
- **Languages:** English, Telugu, Hindi